RSC Advances

RSC Publishing

PAPER

Imaging liquid crystal defects

Ingo Dierking* and Paul Archer

Cite this: RSC Adv., 2013, 3, 26433

Received 26th September 2013
Accepted 17th October 2013

DOI: 10.1039/c3ra45390a

www.rsc.org/advances

Polymer stabilization is used to image the director fields of a range of different $s = \pm 1$ and $s = \pm 1/2$ Schlieren and umbilical defects in nematic liquid crystals, such as radial, vortex and swirl defects of strength $s = +1$. We demonstrate experimentally the director field around such defects and show that they coincide with schematic representations commonly found in literature. The proposed method can be used non-invasively or invasively, depending on the resolution required.

I. Introduction

Defects play a fundamental role in all condensed matter materials, determining their strength, their usefulness and efficiency as functional materials, the quality of their optical properties, their electrical conductivity.[1] The list could easily be extended. Liquid crystals (LC)[2] are materials which display self-organisation, thus the formation of ordered structures without external stimuli. Nevertheless, like in all condensed matter systems, this is generally not achieved without the formation of defects. With liquid crystals being one of the classes of condensed matter materials which are termed as being "soft matter",[3] these defects are readily visible. The reason is that their elastic constants are several orders of magnitude smaller than those of true solids. This implies that the defect extensions are not on the order of several atomic distances, but rather in the range of tens of micrometers. They are thus in principle visible in polarised light microscopy. Fig. 1 shows an exemplary texture of a nematic liquid crystal with so called Schlieren defects. These have a strength s, and a sign. The strength s is defined by the number of brushes, divided by four. The sign depends on the director configuration $n(r)$, the spatial and orientational distribution of the long axis of elongated mesogens, and is determined through rotation of the sample between crossed polarizers; positive, when brushes rotate clockwise, negative when anti-clockwise.

Besides the very interesting overlap of some liquid crystal defects with those of solid state materials, such as edge- and screw-dislocations, liquid crystals are also seen as materials that provide the defect topology to give experimental evidence for the behaviour of defects in the early stages of the universe. This concerns phase transitions from uniform to more ordered states (Kibble)[4] at the very early stages of the universe, to possible analogies in the solid state world (Zurek).[5] Defect annihilation after such a phase transition is certainly also a topic of great interest, especially with respect to scaling laws.[6,7]

Liquid crystal nematic Schlieren and umbilical textures are long known, with many schematic drawings having been published in the past. But surprisingly, rather few attempts have been made to actually visualize $n(r)$ the corresponding liquid crystal structure or director field, *i.e.* the field imaging the long molecular axis. One of them is based on a microprecipitate[8] (imaging a $s = -1$ defect), another on bubble decoration[9] (imaging a pair of $s = -1$ and $s = +1$ swirl defects). Both of these methods have the disadvantage that experiments have to be done at an additional LC–air interface. Recently, a method based on confocal microscopy has been employed in the study of director fields, Fluorescent Confocal Polarized Microscopy (FCPM,[10]). This indeed allows imaging of director fields even in three dimensions, but is experimentally quite advanced with

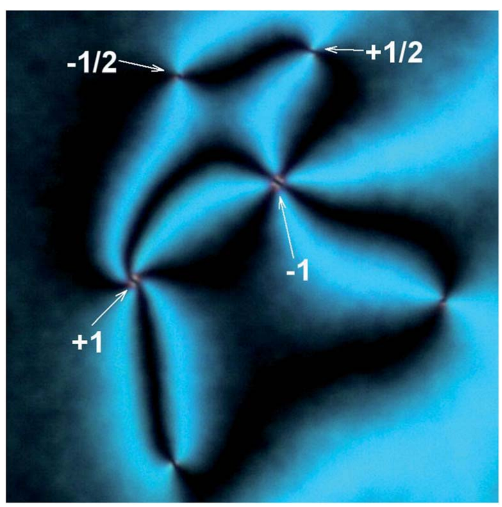

Fig. 1 Polarizing microscopic image of a nematic liquid crystal Schlieren texture with defects of different strength and sign, $s = \pm 1$ and $s = \pm 1/2$, as indicated. (E7 + 5% RM257 monomer, planar boundary conditions, defects obtained by a temperature quench across the isotropic to nematic transition.)

School of Physics and Astronomy, University of Manchester, Oxford Road, Manchester M13 9PL, UK. E-mail: ingo.dierking@manchester.ac.uk

only few apparatus available for liquid crystal research worldwide, thus not readily accessible. It should be noted that also transmission electron microscopy (TEM) has been employed in the study of polymer network structures.[11] Both SEM and TEM have their advantages and disadvantages. TEM sample preparation is relatively complicated with processes of staining, embedding and ultramicrotomy involved. The advantage is that it allows to observe the internal structure through the sample, while SEM is generally more based on imaging the surface. (To produce an image of the three dimensional network structure through the sample, also SEM needs to rely on elaborate sample preparation, which includes embedding and oblique cuts through the cell.[12]) TEM in principle allows a larger magnification than SEM, but nevertheless seems to provide slightly less network detail, possibly due to the smaller depth of field. The network features of polymer stabilised liquid crystals are on average of the order of 0.1 μm. This is easily achieved by SEM and does not need the higher resolution of a TEM. Furthermore, due to their three dimensional nature, SEM images need less interpretation than the two dimensional TEM pictures.

II. Experimental

For the study, commercial liquid crystal mixtures from Merck, Darmstadt, were used, E7 with a positive and ZLI-2806 with a negative dielectric anisotropy. The bifunctional photoreactive monomer was in all cases the standard RM257, also from Merck, at a concentration of 5% by weight. A small amount (2% of the monomer concentration) of benzoin methyl ether (BME) was added to initiate the photopolymerisation. This was exclusively carried out by UV irradiation for two hours with a peak wavelength of $\lambda = 352$ nm and a power density of 0.1 mW cm^{-2}. Cells were irradiated directly after defect formation to avoid annihilation processes. Various sandwich cells, commercial (Linkham, thickness 5 μm), as well as house made cells were employed to achieve different boundary conditions, planar in the case of E7 and homeotropic for the negative dielectric ZLI-2806 mixture. Homeotropic cells were produced in house at a thickness of 15 μm and 50 μm. They carried an OTMS alignment layer (trimethoxy-octadecyl-silane, also known as octadecyltrimethoxysilane from Sigma Aldrich). Cells were made by using a mixture of 90% deionized water, 10% methanol and one drop of OTMS. Substrates were dipped into the solution for 20 s and heating to 100 °C for 5 min, followed by baking for 60 min at 150 °C.

Schlieren and umbilic textures were initiated in planar cells through temperatures quenches across the isotropic to nematic transition and by electric field application in homeotropic cells with a negative dielectric material, respectively. They were observed in a polarized microscope (Leica DMLP), equipped with a Linkam hotstage and temperature controller (Mettler FP82HT and FP90) for control of relative temperatures to ±0.1 K. For the induction of umbilic defects, electric fields were applied by a Thurlby Thandar TGA12101 function generator in combination with an in house built power amplifier. Digital texture images were captured with an IDS UI-5460-C camera, at 1200 × 800 pixels resolution. After polymerization, the liquid crystal was removed by a suitable solvent (for example hexane, dichloromethane or ether) and the polymer film dried at slightly elevated temperatures of 30 °C. The polymer surface was imaged by electron scanning microscopy (Zeiss EVO60) with samples covered by a 2 nm layer of chromium, followed by addition of a 8 nm layer of gold.

III. Results and discussion

In this paper we visualise all of the nematic liquid crystal defects with $s = +1$, $s = -1$, $s = +1/2$, $s = -1/2$. This is done by polymer stabilization.[13] A bifunctional elongated monomer is mixed with the thermotropic, calamitic liquid crystal. It adopts the orientation of the director field of its host, and is then polymerised *via* UV illumination. The formed polymer network is thus templating the liquid crystal director field in which it was formed. Its structure can be studied in several different ways.

One can maintain the bi-continuous material of polymer network and liquid crystal, heat slightly above the point where

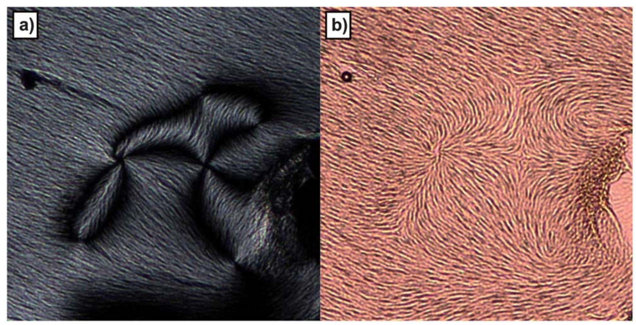

Fig. 2 Imaging of the defect director field through (a) the residual birefringence of the LC and (b) by reflection microscopy. Both are non-invasive methods. The images show the same sample area of E7 after polymerisation with 5% RM257 with a $s = +1$ defect connecting with a $s = -1$ one. Defects were obtained by a temperature quench of a cell with planar boundary conditions.

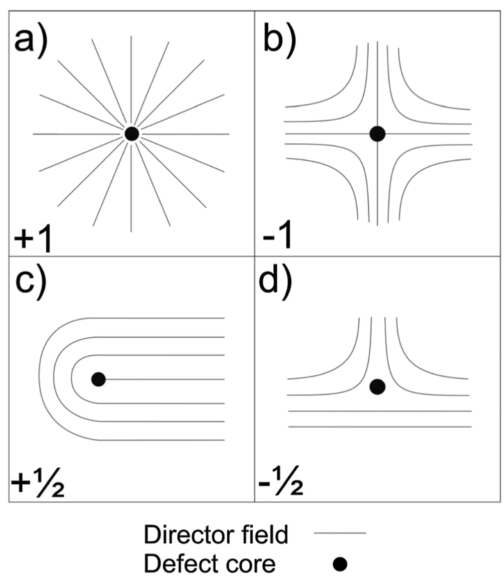

Fig. 3 Schematic illustration of the commonly observed Schlieren defects of strength and sign $s = \pm1$ and $s = \pm1/2$.

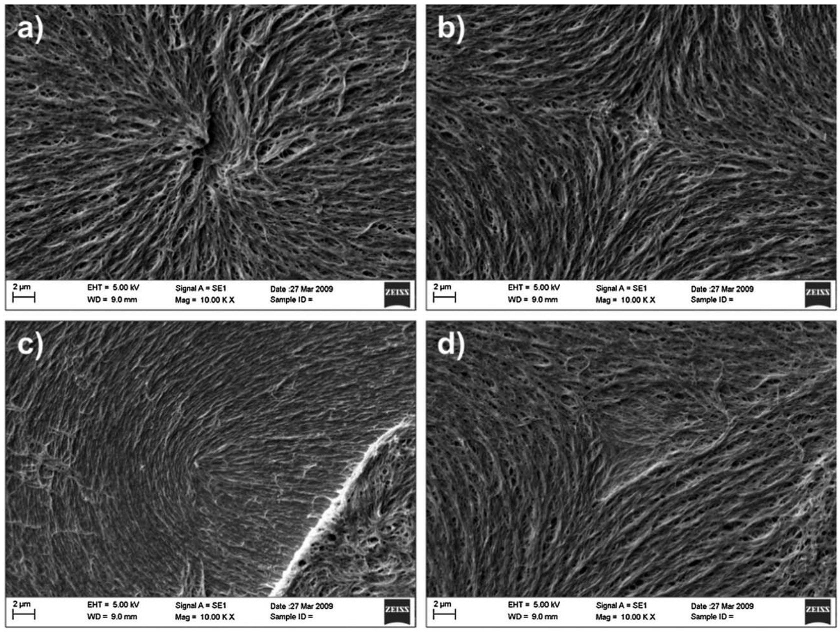

Fig. 4 Direct experimental (SEM) images of the polymer templated director fields schematically shown in Fig. 3. The imaged defects were obtained by a temperature quench of E7 in cells with planar boundary conditions.

the LC becomes isotropic, and image the polymer network in an optical microscope through the residual birefringence, caused by interactions between the LC molecules and the polymer network. This is a non-invasive technique as shown in Fig. 2(a). We image the liquid crystalline director field exhibiting all the features of $s = \pm 1$ and $s = \pm 1/2$ defects. In Fig. 2(b) this can even better be seen by reflection microscopy of thin metal coated samples, showing the same area as produced in Fig. 2(a).

Defects can indeed be imaged to even better resolution by polymer stabilization and subsequent scanning electron microscopy (SEM), an invasive technique, where the liquid crystal is removed by a solvent after polymer formation. Fig. 3 schematically illustrates the commonly observed Schlieren defects of strength and sign $s = \pm 1$ and $s = \pm 1/2$, while Fig. 4 shows the corresponding SEMs of the actual polymer network. These defects were produced in planar cells during a temperature quench.

The application of an electric field to a negative dielectric nematic confined in a cell with homeotropic boundary conditions leads to the formation of umbillical defects. These all have a strength of either $s = +1$ or $s = -1$, with no half integer defects occurring. Different $s = +1$ defects can be observed, namely radial and vortex defects, which are schematically shown in Fig. 3(a) and 5(a), respectively. Both of these defects have the same sign and strength. They can not be distinguished by polarizing microscopy, in contrast to the $s = -1$ defect shown in Fig. 5(c). The corresponding direct images of these defects are shown in Fig. 4(a) and 6(a).

Fig. 6(d) shows a $s = -1$ defect (left) together with a $s = +1$ vortex defect (right). In the non-polymerized state, due to the liquid and elastic nature of the LC, there acts an attractive force between these defects, they would approach each other and annihilate on contact to form a uniform director field.

Electric field induced umbilical defects are not strictly singularities, but instead exhibit a finite defect core, which is influenced by externally applied electric fields as demonstrated by polarizing microscopy in Fig. 7. The behaviour of the core

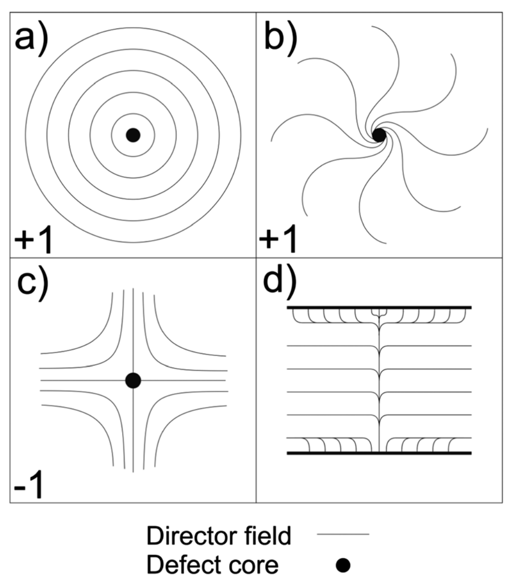

Fig. 5 Schematic illustration of $s = +1$ defects which have been proposed in the past (a) vortex defect and (b) swirl defect. These can not be distinguished from each other by polarizing microscopy, neither from the $s = +1$ defect shown in Fig. 3(a), in contrast to the $s = -1$ defect of part (c). (d) illustrates the finite core size of umbilical defects. These are thus not singularities in the strict sense.

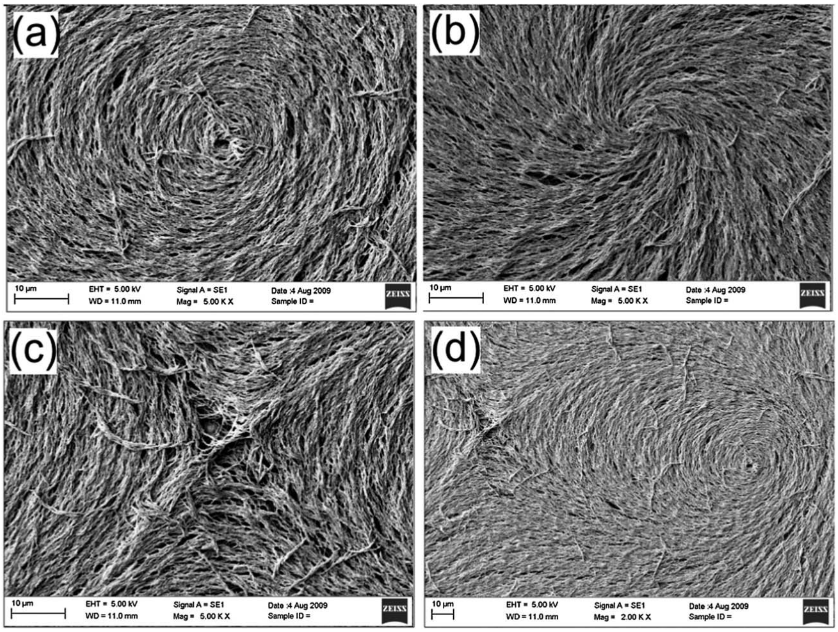

Fig. 6 Direct images of (a) the vortex defect and (b) the swirl defect, which are both $s = +1$ defects. These can not be distinguished by normal polarizing microscopy. In contrast, the $s = -1$ defect of part (c) can be distinguished from the above two. Part (d) shows a pair of $s = -1$ (left) and $s = +1$ vortex defect (right). The defects were obtained by electric field application to the negative dielectric nematic ZLI-2806 under homeotropic boundary conditions.

size can also be qualitatively estimated from the SEMs of the polymer stabilized defects as shown in Fig. 8 for (a) a $s = -1$ and (b) a $s = +1$ swirl defect. The images do not directly show the finite core, as this would be too small, with the polymer network strands being wider than the core itself. But at the location of the core a void which changes with applied field amplitude can be imaged. It is anticipated that these voids result in regions with large director curvature where the polymer is expelled during polymerization, a behaviour which we have observed also in other systems, like the oily streaks regions of cholesterics and defect regions between smectic layers in TGB phases.

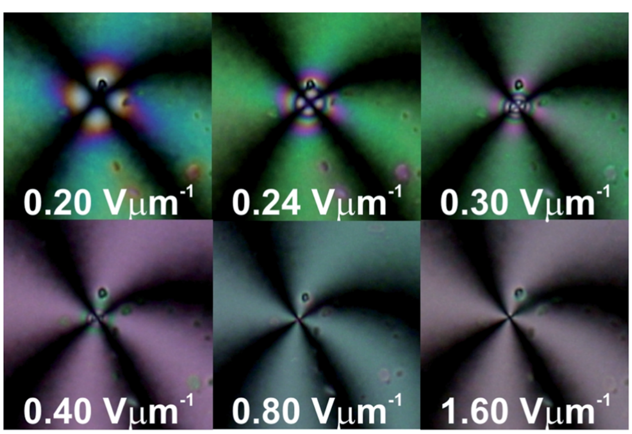

Fig. 7 Electric field induced umbilical defect of negative dielectric nematic ZLI-2806 confined to a cell with homeotropic boundary conditions as the applied amplitude is increased (individual picture size 30 × 30 μm).

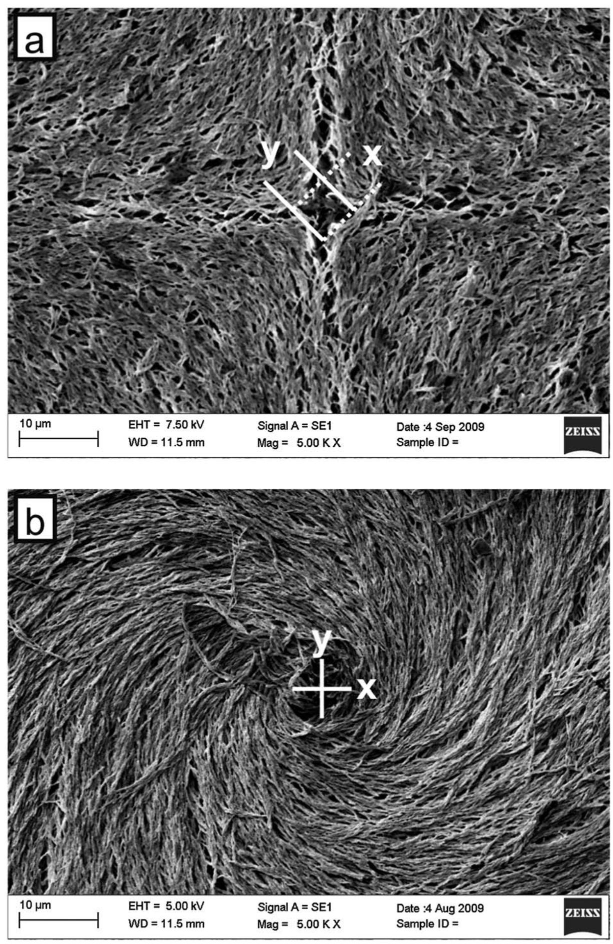

Fig. 8 The core of (a) a $s = -1$ and (b) a $s = +1$ swirl defect, directly imaged from the polymer network.

IV. Conclusions

In conclusion, we have demonstrated a simple and reliable method for the direct imaging of LC defects, which can be used both non-invasively and invasively, depending on the resolution needed. Director fields $n(r)$ of nematic defects were experimentally illustrated for a range of $s = \pm 1$ and $s = \pm 1/2$ defects.

References

1 M. Kleman, *Points, Lines and Walls in Liquid Crystals, Magnetic Systems and Various Ordered Media*, Wiley, Chichester, 1983.
2 S. Chandrasekhar, *Liquid Crystals*, Cambridge University Press, Cambridge, 2nd edn, 1992.
3 M. Kleman and O. D. Lavrentovich, *Soft Matter Physics: An Introduction*, Springer, New York, 2003.
4 T. Kibble, *J. Phys. A: Math. Gen.*, 1976, **9**, 1387.
5 W. H. Zurek, *Nature*, 1985, **317**, 505.
6 I. Chuang, R. Durrer, N. Turok and B. Yurke, *Science*, 1991, **251**, 1336; I. Chuang, N. Turok and B. Yurke, *Phys. Rev. Lett.*, 1991, **66**, 2472.
7 I. Dierking, O. Marshall, J. Wright and N. Bulleid, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2005, **71**, 061709; I. Dierking, M. Ravnik, E. Lark, J. Healey, G. P. Alexander and J. M. Yeomans, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2012, **85**, 021703.
8 J. Rault, *C. R. Acad. Sci.*, 1971, **272**, 1275.
9 P. Cladis, M. Kleman and P. Pieranski, *C. R. Seances Acad. Sci., Ser. B*, 1971, **273**, 275.
10 I. I. Smalyukh, *Mol. Cryst. Liq. Cryst.*, 2007, **477**, 23.
11 M. Brittin, G. R. Mitchell and A. S. Vaughan, *Liq. Cryst.*, 2000, **27**, 693; M. Brittin, G. R. Mitchell and A. S. Vaughan, *J. Mater. Sci.*, 2001, **36**, 4911.
12 I. Dierking, L. L. Kosbar, A. Afzali-Ardakani, A. C. Lowe and G. A. Held, *J. Appl. Phys.*, 1997, **81**, 3007.
13 I. Dierking, *Adv. Mater.*, 2000, **12**, 167.